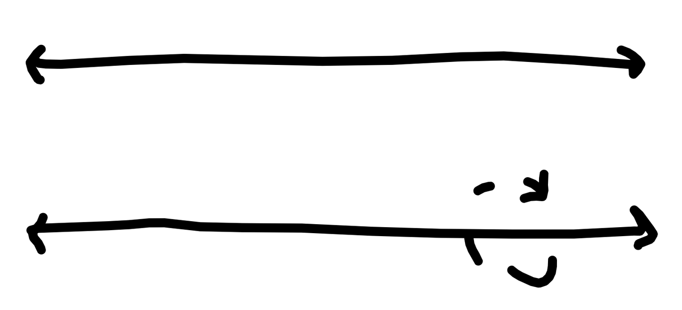
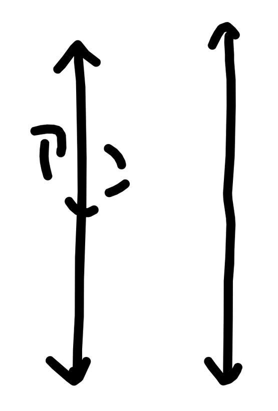
Volume Using Shells

Section 7.3

1 - 20 - 26

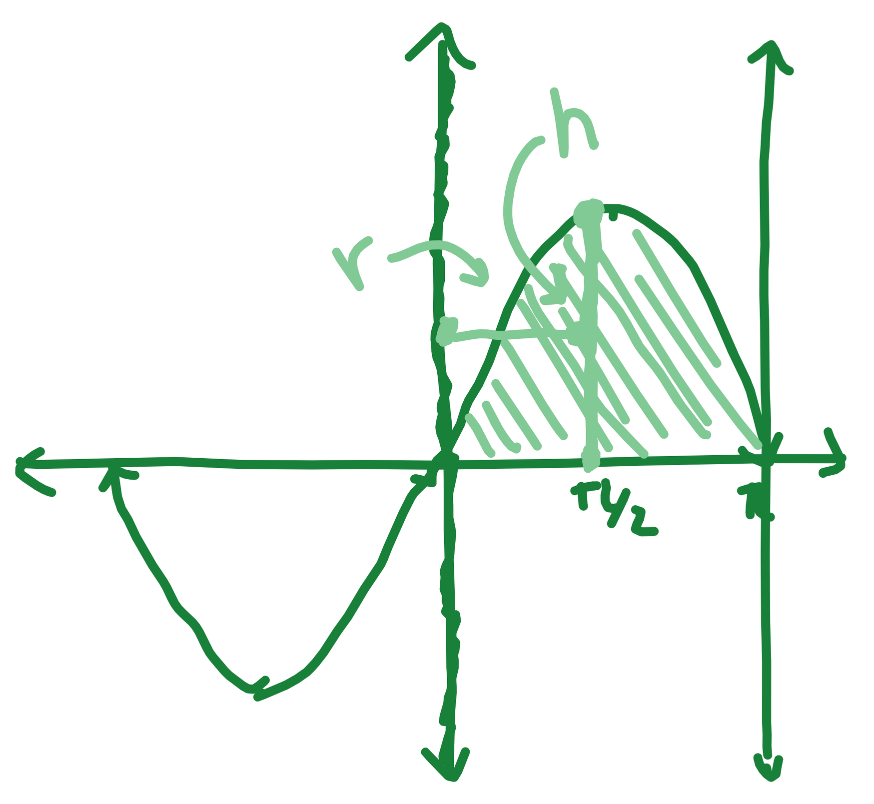
A) Line Rotations

Horizontal (y=a)	Vertical (x=b)
	
$2\pi\int_c^d rh\,dy$	$2\pi\int_a^b rh\,dx$

B) Base

Ex:

$$\begin{cases} y=\sin x \\ x=\pi \\ y=0 \end{cases}$$



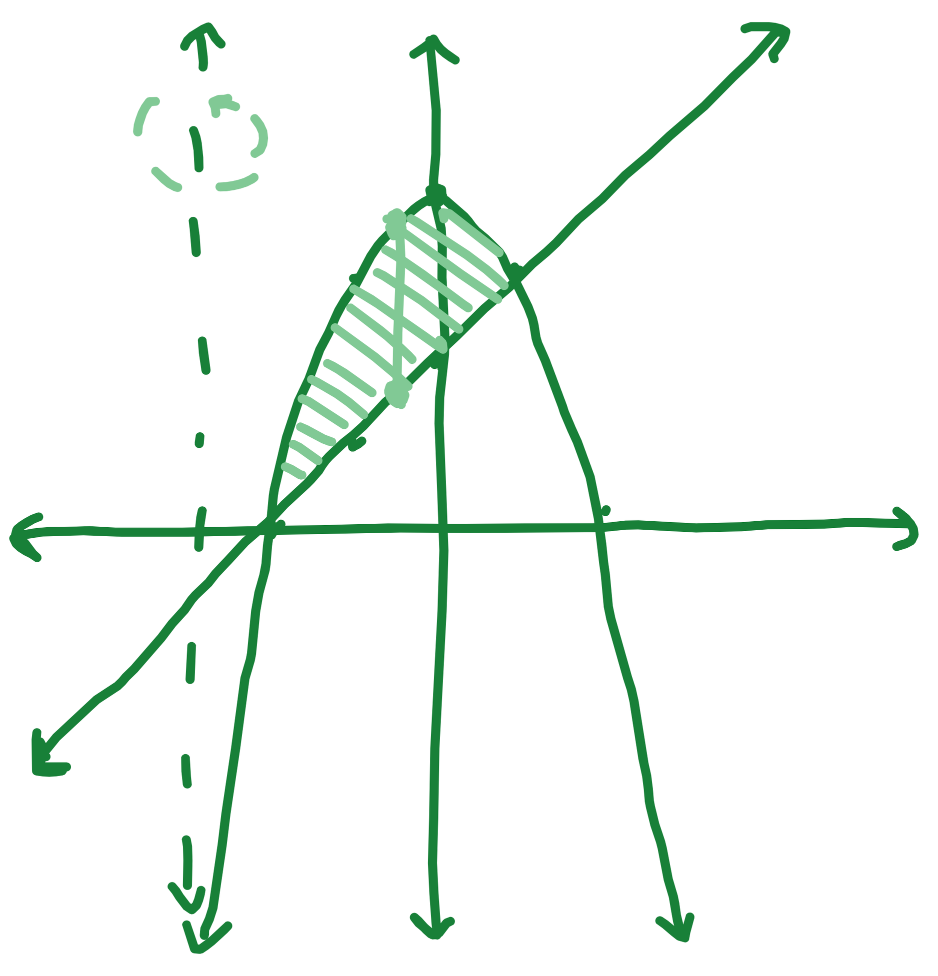
Rotate around

y-axis

$$\boxed{2\pi\int_0^{\pi}(x-0)(\sin x-0)\,dx}$$

Ex:

$$\begin{cases} y=4-x^2 \\ y=x+2 \end{cases}$$



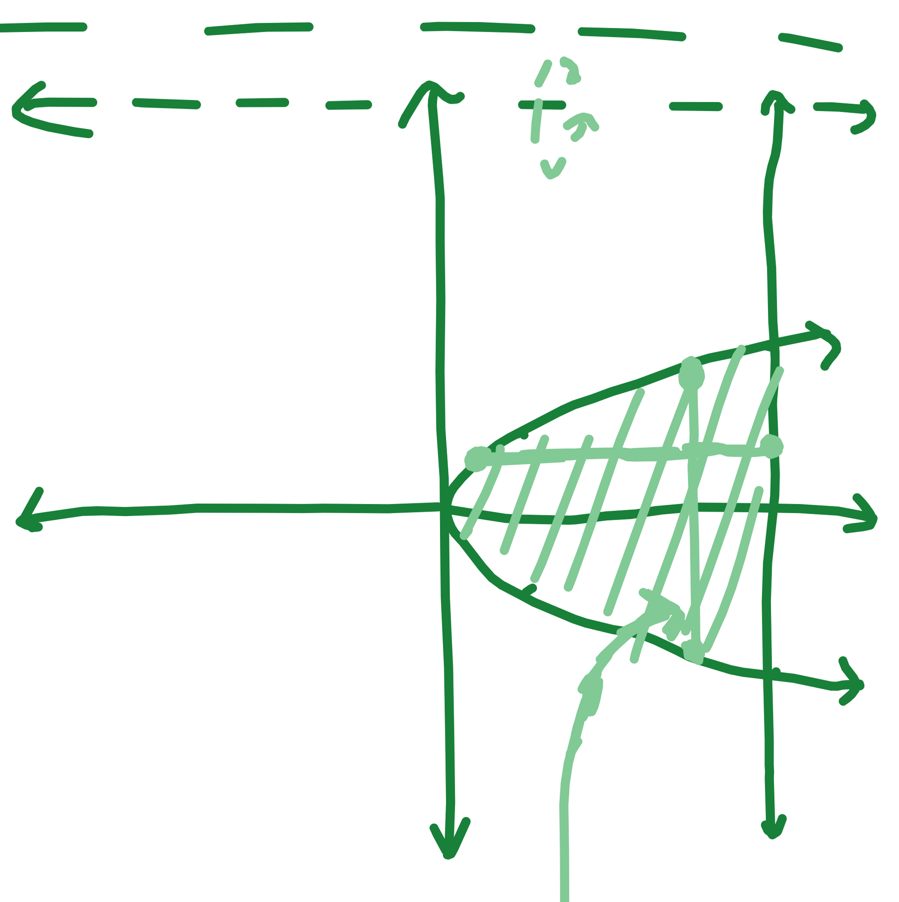
Rotate around

x=-3

$$\boxed{2\pi\int(x+3)\left(-x^2-x+2\right)dx}$$

Ex:

$$\begin{cases} x=y^2 \\ x=4 \end{cases}$$



rotate around

y=5

a) dy	b) dx
$2\pi\int_{-2}^{2}(5-y)(4-y^2)\,dy$	$\pi\int_0^4 (5+\sqrt{x})^2-(5-\sqrt{x})^2\,dx$